4E1306

Roll No. [redacted]

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B. Tech. IV - Sem. (Main / Back) Exam., - 2025
Artificial Intelligence and Data Science
4AID4-06 Theory of Computation
CS, IT, AID, CAI, CCS, CDS, CIT

Time: 3 Hours **Maximum Marks: 70**

Instructions to Candidates:

Attempt all ten questions from Part A, five questions out of seven questions from Part B and three questions out of five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/ calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No. 205)

1. NIL
2. NIL

PART – A

(Answer should be given up to 25 words only) **[10×2=20]**

All questions are compulsory

Q.1 What are acceptors and transducer in context of Turing machine?

Q.2 Define a Finite Automata.

Q.3 Write applications of Mealy and Moore machines.

Q.4 What are the computable functions?

Q.5 Define any one closure property of regular sets.

Q.6 Show that the grammar $S \rightarrow aB \mid ab$, $A \rightarrow aAB \mid a$, $B \rightarrow Abb \mid b$ is ambiguous.

Q.7 Describe the context-sensitive grammar (CSG).

Q.8 What is NP complete and NP hard problem?

Q.9 What is the use of Null Production and Unit Production?

Q.10 What is recursively enumerable language?

PART - B

[5×4=20]

(Analytical/Problem solving questions)

Attempt any five questions

Q.1 Describe the terms tractable problem, intractable problem and undecidability problem in context of computation.

Q.2 Discuss Turing machine as language acceptors and transducers. Construct Turing machine for language $L = \{a^n b^n c^n \mid n >= 1\}$.

Q.3 What is the role of CFG for PDA? Construct the PDA for language $L = \{a^n b^{2n} \mid n >= 1\}$.

Q.4 Distinguish CNF and GNF with examples. Also explain problem related to them.

Q.5 Explain the Chomsky classification with the suitable example of each grammar.

Q.6 Discuss the equivalence of DFA and NDFA.

Q.7 Construct the CSG for the language $L = \{a^n b^n c^n \mid n >= 1\}$.

PART – C

(Descriptive/Analytical/Problem Solving/Design Questions)

Attempt any three questions

Q.1 Elaborate pumping lemma for regular language. Show that the language $L = \{a^n b^n \mid n > 0\}$ is not regular by using pumping lemma.

Q.2 Deduce Myhill-Nerode theorem and prove that following language is regular: $L = \{w \in \{a, b\}^* \mid w$ having even number of a's$\}$.

Q.3 Explain conversion of Finite automata to regular expression and vice versa. Prove that the strings recognized by the following finite automata are: $(a + a(b + aa)^*b)^*a(b + aa)^*a)$. Where P is the initial state and R is the final state.

Current State	Current Input: a	Current Input: b
P	P, Q	
Q	R	P, Q
R	Q	

Q.4 Discuss the solution of travelling salesman problem with a real-world example.

Q.5 Explain the following terms with examples -

(1) Universal Turing Machine

(2) Multitrack Turing Machine
